Robust Sigma-Point Filtering for Nonlinear Systems with Non-Additive Noise

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***Abstract*— This paper addresses the problem of robust state estimation for nonlinear systems with non-additive noise. We build on ideas related to our previous work [1] by augmenting the sigma points with noise variables and embedding them into a minimax formulation. Specifically, the objective function is the variance of the state estimation error, the minimizer corresponds to the robust filter, and the maximizer represents the least favorable model within an ambiguity set about the nominal model. Simulation results demonstrate that the proposed filter achieves superior robustness and estimation accuracy compared to existing robust and standard sigma-point filters.**

I. Introduction

State estimation plays a fundamental role in modern control and signal processing applications, such as autonomous navigation [2], [3], target tracking [4], sensor fusion [5], and learning brain networks (DCM modeling) [6]. Among various estimation techniques, the Kalman filter (KF) and its nonlinear extensions have been widely adopted due to their recursive structure and computational efficiency [7]. For nonlinear systems, sigma-point Kalman filtering methods [8], [9], including the unscented Kalman filter (UKF) [10] and the cubature Kalman filter (CKF) [11], achieve improved estimation accuracy by deterministically sampling sigma-points to approximate the propagation of probability densities [12].

In a broader context, robust state estimators have been extensively studied in the estimation and control literature to ensure reliable performance under uncertainty and outlier contamination, see for instance [13], [14]. The performance of conventional sigma-point filters, however, can deteriorate severely in the presence of such model inaccuracies. To address this issue, we recently proposed a robust sigma-point filtering approach to reduce the sensitivity of estimation errors to modeling mismatches, [1]. More precisely, we developed a robust sigma-point filtering framework formulated as a minimax game, formerly introduced in the linear setting in [15]–[21], between two players: the state estimator, which aims to minimize the variance of the estimation error, and the nature, which selects the least favorable model within an ambiguity set centered around the nominal model to maximize this error.

The simplest instance of the sigma-point Kalman filter considers state space models with additive noise, where the process and measurement noises enter linearly in the model. This assumption is often violated in practical systems, such as aided inertial navigation [2], nonlinear measurement sensors [22], and stochastic systems with multiplicative noise [23], where the noises are non-additive and interact with the system dynamics in a nonlinear manner. Under such conditions, the standard sigma-point propagation rules fail to accurately capture the impact of the noise, leading to degraded estimation performance.

In this paper, we focus on nonlinear systems with non-additive noises and develop a robust sigma-point filtering approach. Specifically, we consider the minimax robust filtering problem mentioned above and approximate the nominal model, which is the center of the ambiguity set, in the same way as in the sigma-point Kalman filter. This allows us to characterize the minimizer of the minimax game, which represents the proposed robust sigma-point Kalman filter. Simulation results demonstrate that the proposed filter achieves superior robustness and estimation accuracy compared with both conventional and existing robust sigma-point filters. We emphasize that, although this work builds on ideas related to [1], it does not address the characterization of the least favorable model nor the generation of data from it, which are beyond the scope of this paper.

The remainder of this paper is organized as follows. Section II reviews the robust sigma-point filtering framework for systems with non-additive noise. In Section III we derive the robust sigma-point filter for state space models with the non-additive noise. Section IV presents some simulation results validating the effectiveness of the proposed robust filter. Finally, Section V concludes the paper.

Notation. $p(z) \sim \mathcal{N}(\mu, P)$ means that the probability density $p(z)$ is Gaussian with mean μ and covariance matrix P. Given a symmetric matrix P, $P > 0$ and $P \geq 0$ mean that P is positive definite and semi-definite, respectively. I_n denotes the identity matrix of dimension n; $A^\top$ denote the transpose of matrix A.

II. Non-additive Sigma-point KF

We consider the nominal discrete-time nonlinear state space model with non-additive noise:

$$\begin{cases} x_{t+1} = f(x_t, v_t) \\ y_t = h(x_t, w_t) \end{cases} \tag{1}$$

where $x_t \in \mathbb{R}^n$ is the state; $y_t \in \mathbb{R}^m$ is the observation; $v_t \in \mathbb{R}^n$ and $w_t \in \mathbb{R}^m$ are independent white Gaussian noises (WGN) with zero mean and covariance matrix equal to $Q \in \mathbb{R}^{n\times n}$ and $R \in \mathbb{R}^{m\times m}$, respectively, which are full rank. Moreover, we assume that v_t and w_t are independent from the initial state x_0.

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An estimate of the state x_t given the observations $Y_t := \{y_s,\ s \leq t\}$ can be obtained by means of the non-additive Sigma-Point Kalman filter, [12, Section 8]. Let $p_t(y_t, x_t|Y_{t-1})$ denote the conditional density of y_t and x_t given Y_{t-1} according to model (1). The main idea in this estimation paradigm is to approximate $p_t(y_t, x_t|Y_{t-1})$ by a Gaussian conditional density $\bar{p}_t(y_t, x_t|Y_{t-1}) \sim \mathcal{N}(m_t, K_t)$ whose mean and covariance matrix are computed from the propagated sigma-points. The latter are generated over the augmented state-noise space in order to capture the coupling between state and noise in the nonlinear transformations (see below).

Notation. *Consider two independent random variables $x \sim \mathcal{N}(\hat{x}, P)$ and $u \sim \mathcal{N}(0, U)$, corresponding to the state and the noise, respectively. Then, (augmented) sigma-points are denoted by*

$$(\mathcal{X}^{i,x},\ \mathcal{X}^{i,u}) = \sigma_i(\hat{x}, P, U),$$

with $i = 1, \ldots, p$, where $\mathcal{X}^{i,x}$ and $\mathcal{X}^{i,u}$ represent the sigma-points corresponding to x and u, respectively. Finally, W_m^i and W_c^i,with $i = 1, \ldots, p$, denote the weights with respect to the mean and covariance matrix, respectively.

There are many ways to define the sigma-points and the weights (see [12, Sec. 5-7]). For example, according to the unscented transformation, we have

$$\begin{bmatrix} \mathcal{X}^{i,x} \\ \mathcal{X}^{i,u} \end{bmatrix} = \begin{cases} \bar{x} + \sqrt{\lambda + n_a}(\sqrt{\bar{P}})_i, & \text{if } 1 \leq i \leq n_a \\ \bar{x} - \sqrt{\lambda + n_a}(\sqrt{\bar{P}})_{i-n_a}, & \\ & \text{if } n_a + 1 \leq i \leq 2n_a \\ \bar{x}, & \text{if } i = 2n_a + 1 \end{cases}$$

where n_a is is the sum of the vector dimensions of x and u,

$$\bar{x} := \begin{pmatrix} \hat{x} \\ 0 \end{pmatrix}, \quad \bar{P} := \begin{pmatrix} P & 0 \\ 0 & U \end{pmatrix}$$

and $(\sqrt{\bar{P}})_i \in \mathbb{R}^{n_a}$ is the i-th column of $\sqrt{\bar{P}}$ which is a square root matrix of $\bar{P}$. The corresponding weights are

$$\begin{aligned} W_m^i &= \frac{\lambda}{n_a + \lambda}, \quad W_c^i = W_m^i + 1 - a^2 + b, && \text{if } i = 2n_a + 1, \\ W_m^i &= W_c^i = \frac{1}{2(n_a + \lambda)}, && \text{if } 1 \leq i \leq 2n_a \end{aligned} \tag{2}$$

where $\lambda = a^2(n_a + \kappa) - n_a$ and parameters a, b, κ can be chosen as explained in [12]. If we consider the spherical cubature transformation, we have

$$\begin{bmatrix} \mathcal{X}^{i,x} \\ \mathcal{X}^{i,u} \end{bmatrix} = \begin{cases} \bar{x} + \sqrt{n_a}(\sqrt{\bar{P}})_i, & \text{if } 1 \leq i \leq n_a \\ \bar{x} - \sqrt{n_a}(\sqrt{\bar{P}})_{i-n_a}, & \text{if } n_a + 1 \leq i \leq 2n_a, \end{cases}$$

and the corresponding weights are $W_c^i = W_m^i = 1/2n_a$.

III. Robust non-additive Sigma-Point Kalman Filter

We consider the situation in which the actual model deviates from the nominal model (1). Our goal is to develop a non-additive sigma-point Kalman filter which is robust to the fact that the actual and the nominal model are not exactly the same. We consider the setup originally proposed in [24] and later extended to the nonlinear case with additive noise in [1, Section V]. In this framework, the estimation problem at time t is formulated as a dynamic minimax game between two players: the state estimator, which minimizes the variance of the state estimation error, and the nature, which selects the conditional density of y_t given x_t that maximizes this error within an ambiguity set around the nominal conditional density.

Let $\psi_t(y_t|x_t)$ and $\tilde{\psi}_t(y_t|x_t)$ denote the conditional density of y_t given x_t corresponding to the nominal and the actual model, respectively. The ambiguity set is defined as

$$\mathcal{B}_t := \left\{ \tilde{\psi}_t(y_t|x_t) \text{ s.t. } \mathcal{D}(\tilde{\psi}_t, \psi_t) \leq c \right\} \tag{3}$$

where

$$\mathcal{D}(\tilde{\psi}_t, \psi_t) := \iint \tilde{\psi}_t\left(y_t|x_t\right) \tilde{p}_t\left(x_t|Y_{t-1}\right) \ln\left(\frac{\tilde{\psi}_t\left(y_t|x_t\right)}{\psi_t\left(y_t|x_t\right)}\right) dy_t dx_t$$

is the conditional Kullback–Leibler (KL) divergence used to measure the deviation between $\tilde{\psi}_t$ and ψ_t; $\tilde{p}_t(x_t|Y_{t-1})$ denotes the actual a priori conditional density of x_t given Y_{t-1}; finally, $c \geq 0$ represents the radius of the ball $\mathcal{B}_t$ and is called tolerance. In plain words, we fix a budget of uncertainty at each time step, ensuring that the uncertainty is not concentrated at specific time instants where the estimator is more vulnerable. In this way, the robust estimator is characterized by the following dynamic minimax game:

$$(\tilde{\psi}_t^\star, g_t^\star) = \arg \min_{g_t \in \mathcal{G}_t} \max_{\tilde{\psi}_t \in \mathcal{B}_t} J_t(\tilde{\psi}_t, g_t) \tag{4}$$

where

$$J_t(\tilde{\psi}_t, g_t) = \frac{1}{2} \iint \left\| x_t - g_t\left(y_t\right) \right\|^2 \tilde{\psi}_t(z_t|x_t) \tilde{p}_t\left(x_t|Y_{t-1}\right) dz_t dx_t \tag{5}$$

denotes the variance of the estimation error under the actual model; $\mathcal{G}_t$ denotes the class of estimators with finite second-order moments with respect to $\tilde{\psi}_t \in \mathcal{B}_t$. Finally, $\tilde{\psi}_t$ must satisfy the constraint:

$$\iint \tilde{\psi}_t\left(y_t|x_t\right) \tilde{p}_t\left(y_t|Y_{t-1}\right) dy_t dx_t = 1. \tag{6}$$

Since both $\mathcal{B}_t$ and $\mathcal{G}_t$ are convex and compact, and the objective function J_t is linear in $\tilde{\psi}_t$ and convex in g_t, there exists a saddle point $(\tilde{\psi}_t^\star,\ g_t^\star)$ which solves (4).

Let $w_t := [\, x_t^\top \ y_t^\top \,]^\top$ and

$$\begin{aligned} p_t\left(w_t|Y_{t-1}\right) &:= \psi_t(y_t|x_t) \tilde{p}_t\left(x_t|Y_{t-1}\right), \\ \tilde{p}_t\left(w_t|Y_{t-1}\right) &:= \tilde{\psi}_t^\circ(y_t|x_t) \tilde{p}_t\left(x_t|Y_{t-1}\right), \end{aligned} \tag{7}$$

where

$$\tilde{\psi}_t^\circ(y_t|x_t) = \frac{1}{M_t} \exp\left(\frac{\theta_t}{2} \left\| x_t - g_t(y_t) \right\|^2\right) \psi_t(y_t|x_t), \tag{8}$$

$\theta_t > 0$ is the unique solution to the equation $\mathcal{D}(\tilde{\psi}_t^\circ, \psi_t) = c$, and the normalizing constant is

$$M_t = \iint \exp\left(\frac{\theta_t}{2} \|x_t - g_t(y_t)\|^2\right) \psi_t \tilde{p}_t\left(x_t|Y_{t-1}\right) dy_t dx_t.$$

In [1], it was shown that a saddle point solution to (4) can be obtained by considering the minimax game

$$\min_{g_t \in \mathcal{G}_t} \max_{\tilde{p}_t \in \bar{\mathcal{B}}_t} \bar{J}_t(\tilde{p}_t, g_t) \tag{9}$$

where

$$\bar{J}_t(\tilde{p}_t, g_t) := \frac{1}{2} \int \|x_t - g_t(y_t)\|^2 \, \tilde{p}_t(w_t|Y_{t-1}) dw_t;$$

the new ambiguity set is

$$\tilde{\mathcal{B}}_t := \{\tilde{p}_t(w_t|Y_{t-1}) \text{ s.t. } \mathcal{D}(\tilde{p}_t, p_t) \leq c\}$$

where $\mathcal{D}(\tilde{p}_t, p_t)$ denotes the Kullback-Leibler (KL) divergence between $\tilde{p}_t(w_t|Y_{t-1})$ and $p_t(w_t|Y_{t-1})$. Thus, we can characterize the estimator as the minimizer of (9). However, due to the nonlinearity of the nominal model (1), the distribution $p_t(w_t \mid Y_{t-1})$ is not Gaussian, since the nominal conditional density $\psi_t(y_t \mid x_t)$ is not Gaussian in general. This makes it very difficult to obtain a closed-form solution for the minimax game (9).

To tackle this issue, we consider an approximation of the game in (9) where we substitute $p_t(w_t \mid Y_{t-1})$, i.e. the center of the ambiguity set $\tilde{\mathcal{B}}_t$, with a Gaussian approximation $\bar{p}_t(y_t, x_t|Y_{t-1}) \sim \mathcal{N}(m_t, K_t)$. More precisely, we consider the same approximation employed in the Sigma-Point Kalman Filter, which we now introduce. Let $\hat{x}_t$ denote the prediction of x_t given Y_{t-1}, and let P_t be the covariance matrix of the prediction error $x_t - \hat{x}_t$. Then,

$$m_t = \begin{bmatrix} \hat{x}_t \\ m_{y_t} \end{bmatrix}, \qquad K_t = \begin{bmatrix} P_t & K_{x_t y_t} \\ K_{x_t y_t}^\top & K_{y_t} \end{bmatrix}. \tag{10}$$

The conditional mean and covariance of y_t given Y_{t-1}, and the conditional cross-covariance of x_t and y_t given Y_{t-1} are obtained using the sigma-point transformations:

$$\begin{aligned} m_{y_t} &= \sum_{i=1}^{n_a} W_m^i h(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w}), \\ K_{y_t} &= \sum_{i=1}^{n_a} W_c^i \big(h(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w}) - m_{y_t}\big) \\ &\qquad \times \big(h(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w}) - m_{y_t}\big)^\top, \\ K_{x_t y_t} &= \sum_{i=1}^{n_a} W_c^i \big(\mathcal{X}_t^{i,x} - \hat{x}_t\big)\big(h(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w}) - m_{y_t}\big)^\top \end{aligned} \tag{11}$$

where the sigma-points are obtained from the distribution of x_t given Y_{t-1} and w_t

$$(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w}) = \sigma_i(\hat{x}_t, P_t, R), \quad i = 1 \dots n_a.$$

Theorem 1: Let $\hat{x}_t$ denote the one-step ahead predictor of x_t and $P_t > 0$ the covariance matrix of the prediction error $x_t - \hat{x}$. Consider the approximation of the minimax game (9) in which the ambiguity set $\tilde{\mathcal{B}}_t$ is replaced by

$$\tilde{\mathcal{B}}_t^a := \{\tilde{p}_t(w_t|Y_{t-1}) \text{ s.t. } \mathcal{D}(\tilde{p}_t, \bar{p}_t) \leq c\}.$$

Algorithm 1 Non-additive Sigma-Point Robust Kalman filter

Input $\hat{x}_t$, P_t, y_t
Output $\hat{x}_{t+1}$, P_{t+1}

$(\mathcal{X}_t^{i,x},\ \mathcal{X}_t^{i,w}) = \sigma_i(\hat{x}_t, P_t, R)$
$\mathcal{Y}_t^i = h(\mathcal{X}_t^{i,x}, \mathcal{X}_t^{i,w})$
$m_{y_t} = \sum_i W_m^i \mathcal{Y}_t^i$
$K_{y_t} = \sum_i W_c^i (\mathcal{Y}_t^i - m_{y_t})(\mathcal{Y}_t^i - m_{y_t})^\top$
$K_{x_t y_t} = \sum_i W_c^i (\mathcal{X}_t^{i,x} - \hat{x}_t)(\mathcal{Y}_t^i - m_{y_t})^\top$
$L_t = K_{x_t y_t} K_{y_t}^{-1}$
$\hat{x}_{t|t} = \hat{x}_t + L_t (y_t - m_{y_t})$
$P_{t|t} = \tilde{P}_t - L_t K_{y_t} L_t^\top$
Find $\theta_{t|t}$ s.t. $\gamma(P_{t|t}, \theta_{t|t}) = c$
$\tilde{P}_{t|t} = (P_{t|t}^{-1} - \theta_{t|t} I)^{-1}$
$(\mathcal{X}_{t|t}^{i,x},\ \mathcal{X}_{t|t}^{i,v}) = \sigma_i(\hat{x}_{t|t}, \tilde{P}_{t|t}, Q)$
$\hat{\mathcal{X}}_{t+1}^i = f(\mathcal{X}_{t|t}^{i,x}, \mathcal{X}_{t|t}^{i,v})$
$\hat{x}_{t+1} = \sum_i W_m^i \hat{\mathcal{X}}_{t+1}^i$
$P_{t+1} = \sum_i W_c^i (\hat{\mathcal{X}}_{t+1}^i - \hat{x}_{t+1})(\hat{\mathcal{X}}_{t+1}^i - \hat{x}_{t+1})^\top$

The robust estimator of x_t given Y_t solving this approximated game is

$$\hat{x}_{t|t} = \hat{x}_t + L_t \left(y_t - m_{y_t}\right), \tag{12}$$

in which the filtering gain is $L_t = K_{x_t y_t} K_{y_t}^{-1}$, where $K_{x_t y_t}$ and K_{y_t} are defined as in (11). The nominal covariance matrix of the estimation error $x_t - \hat{x}_{t|t}$ is

$$P_{t|t} = \tilde{P}_t - L_t K_{y_t} L_t^\top,$$

while the least favorable one is

$$\tilde{P}_{t|t} = (P_{t|t}^{-1} - \theta_{t|t} I)^{-1}. \tag{13}$$

The parameter $\theta_{t|t} > 0$ is called risk sensitivity parameter and is the unique solution to the equation $\gamma(P_{t|t}, \theta_{t|t}) = c$ with

$$\gamma(P, \theta) := \frac{1}{2} \left(\log\det(I_n - \theta P) + \operatorname{tr}\left((I_n - \theta P)^{-1} - I_n\right)\right). \tag{14}$$

Proof: The proof can be found [1]. ■

Since we assume no uncertainty in the transition density of x_{t+1} given x_t, that is uncertainty is only present in the conditional density of y_t given x_t, the prediction step is therefore the same as in the standard non-additive Sigma-Point Kalman Filter. However, the new sigma-points are generated using the least favorable density $\tilde{p}_t(x_t \mid Y_t) \sim \mathcal{N}(\hat{x}_{t|t}, \tilde{P}_{t|t})$ from Theorem 1, with covariance matrix $\tilde{P}_{t|t}$. Algorithm 1 summarizes the resulting robust filter. Notice that, in the limit case $c = 0$ (i.e. the ambiguity set $\tilde{\mathcal{B}}_t^a$ contains only the density $\bar{p}_t(w_t|Y_t)$), we have that the solution to $\gamma(P_{t|t}, \theta_{t|t}) = 0$ is $\theta_{t|t} = 0$ and thus $\tilde{P}_{t|t} = P_{t|t}$. Hence, Algorithm 1 boils down to the standard non-additive Sigma-Point Kalman Filter.

In contrast to the additive noise case considered in [1], the noises v and w are incorporated as additional components of

TABLE I: $\overline{\text{MSE}}$ for the different estimators, with the grey cells highlighting the best performance.

c	RUKF	RUKF-a	RCKF	RCKF-a	N_p	PF
	(UKF)	(UKF-a)	(CKF)	(CKF-a)		
0	0.280	0.347	0.285	0.353	100	0.431
0.0001	0.277	0.347	0.282	0.355	500	0.333
0.001	0.270	0.388	0.274	0.401	1000	0.333
0.005	0.257	0.404	0.261	0.416	2000	0.292
0.01	0.249	0.409	0.253	0.411	3000	0.296
0.015	0.248	0.431	0.249	0.519	5000	0.283
0.02	0.253	0.451	0.248	0.525	8000	0.281
0.03	0.315	0.596	0.259	0.610	10000	0.270
0.05	0.553	0.872	0.352	0.906	20000	0.260

the sigma-point set, i.e. $(\mathcal{X}^{i,x}_{t|t}, \mathcal{X}^{i,v}_{t|t})$ and $(\mathcal{X}^{i,x}_{t}, \mathcal{X}^{i,w}_{t})$. Consequently, the propagated sigma-points inherently account for the noise effects, and the moment matrices K_{y_t}, K_{y_t}, $K_{x_t y_t}$ are obtained directly from the transformed sigma-points, without explicitly involving Q and R.

IV. Simulation experiments

We consider the following nonlinear state space model with non-additive noise:

$$x_{t+1} = \begin{bmatrix} \cos(\alpha(x_{2,t} + v_{2,t})) - 1 + \beta(x_{1,t} + v_{1,t}) + x_{2,t} \\ \gamma\, x_{2,t} + d\, v_{2,t} \end{bmatrix}$$
$$y_t = (x_{2,t} + w_t)^2 - x_{2,t} - x_{1,t}^2 + x_{1,t}. \tag{15}$$

where $x = [x_1^\top,\ x_2^\top]^\top$ and $x_0 \sim \mathcal{N}(\hat{x}_0,\ P_0)$ with $\hat{x}_0 = [1,\ 1]^\top$ and $P_0 = 10^{-3} I_2$; $v = [v_1^\top,\ v_2^\top]^\top$ is the process noise with zero mean and covariance matrix $Q = 0.5 I_2$; w is the measurement noise with zero mean and variance $R = 0.5$. Moreover, the model parameters α, β, γ and d are imprecisely known. Then, a Monte Carlo experiment with $M = 100$ independent runs is conducted. At run k we generate the model parameters which are drawn from the following uniform distributions:

$$\alpha \sim \mathcal{U}(0.3, 0.7), \quad \beta \sim \mathcal{U}(0.3, 0.7),$$
$$\gamma \sim \mathcal{U}(0.8, 0.999), \quad d \sim \mathcal{U}(0.05, 0.1).$$

Then, the corresponding state and measurement trajectories, namely $X_N^k := \{x_1^k \dots x_N^k\}$ and $Y_N^k := \{y_0^k \dots y_N^k\}$, are generated over $N = 50$ discrete time steps. Finally, we estimate the state trajectory from the measurements using the proposed estimators and compare their performance with that of some existing estimators. These numerical experiments have been implemented in MATLAB and executed on a Mechanical Revolution notebook equipped with an AMD R9-7845HX CPU and a GeForce RTX 4070 GPU.

A. Estimators

We consider the robust estimator proposed in Section III using the unscented transformation and spherical cubature transformation. We denote them as RUKF and RCKF, respectively. We compare them with the Unscented Kalman filter in [12], the Cubature Kalman filter in [12] and the bootstrap particle filter, denoted by UKF and CKF, PF, respectively. Moreover, the parameters of the weights for UKF are set as $a = 0.5$, $b = 2$ and $\kappa = 1$. All these estimators are constructed using the (nominal) model of the form in (15) with nominal parameters $\alpha = 0.5$, $\beta = 0.5$, $\gamma = 0.98$, $d = 0.08$ and the noise variances are set equal to the ones of the actual model.

We also consider the robust filters proposed in [25], which were developed for the case of additive noise. In this case as well, we consider the unscented transformation and the spherical cubature transformation, and we denote the corresponding filters as RUKF-a and RCKF-a. It should be noted that these filters cannot be directly applied to the nominal state-space model with non-additive noise of the form (1). To make the problem tractable, we approximate the nonlinear functions f and h in (1) by their first-order Taylor expansions around the noise-free points $v = 0$ and $w = 0$, respectively:

$$f(x_t, v_t) \approx f(x_t, 0) + F_v(x_t)\, v_t,$$
$$h(x_t, w_t) \approx h(x_t, 0) + F_w(x_t)\, w_t,$$

where

$$F_v(x) := \left.\frac{\partial f(x, v)}{\partial v}\right|_{v=0}, \qquad F_w(x) := \left.\frac{\partial h(x, w)}{\partial w}\right|_{w=0}.$$

In practice, the exact Jacobians $F_v(x_t)$ and $F_w(x_t)$ cannot be computed directly because the true state x_t is unknown. Following the same rationale used in the extended Kalman filter linearization, these Jacobians are therefore evaluated at the available state estimates instead. Specifically, we define $F_{v,t} := F_v(\hat{x}_{t|t})$, which is approximated using the filtering estimate $\hat{x}_{t|t}$ (the estimate of x_t given all measurements up to time t), and $F_{w,t} := F_w(\hat{x}_t)$ using the prediction estimate $\hat{x}_t$ (the predictor of x_t given measurements up to time $t-1$). In this setting, we obtain the nominal "additive" model:

$$\begin{cases} x_{t+1} = f_a(x_t) + F_{v,t}\, v_t, \\ y_t = h_a(x_t) + F_{w,t}\, w_t \end{cases} \tag{16}$$

where $f_a(x) := f(x, 0)$ and $h_a(x) := h(x, 0)$. This substitution mirrors the standard EKF implementation, in which the linearized models of $f(\cdot)$ and $h(\cdot)$ are obtained by replacing the unknown true state with its most recent filter or predictor estimate. Finally, we also consider the unscented and the cubature Kalman filters constructed using this additive nominal model and we denote them as UKF-a and CKF-a, respectively[1].

B. Results

The performance of the filters is evaluated using the average mean squared error (MSE) over the time horizon

$$\text{MSE}_t = \frac{1}{M} \sum_{k=1}^{M} \|x_t^k - \hat{x}_{t|t}^k\|^2 \tag{17}$$

[1]Note that, RUKF-a and RCKF-a with c=0 coincide to UKF-a and CKF-a, respectively.

and the average MSE over the run and the time horizon

$$\overline{\mathrm{MSE}} = \frac{1}{N}\sum_{t=1}^{N} \mathrm{MSE}_t, \tag{18}$$

where $\hat{x}^k_{t|t}$ denotes the estimate of x^k_t given $y^k_0 \dots y^k_t$. Table I summarizes the index $\overline{\mathrm{MSE}}$ for the filters, where the robust filters are evaluated for different values of the tolerance c, and the PF is evaluated for different numbers of particles N_p. UKF-a and CKF-a perform significantly worse than UKF and CKF, since the former are constructed using an approximated nominal model with additive noise, while the latter are based on the nominal model with non-additive noise, as in the actual system. As c increases (i.e. as we assume greater model uncertainty), the performance of RUKF-a and RCKF-a worsens. In plain terms, robustness cannot compensate for the parameter uncertainty, since the assumption on the noise model (i.e., additive noise instead of non-additive noise) is incorrect. On the contrary, the proposed estimators RUKF and RCKF outperform the aforementioned filters when the value of c is not too large. However, when c becomes too large, RUKF and RCKF tend to be overly conservative, since robustness excessively reduces the influence of the measurements. Finally, PF exhibits its best performance when N_p is very large, but it is slightly worse than RUKF and RCKF. It is important to point out that RUKF and RCKF are definitely preferable to PF. Indeed, the computational time required to run PF is considerably higher than that of RUKF and RCKF. Let

$$\bar{T} = \frac{1}{M}\sum_{k=1}^{M} T_k, \quad \sigma_T = \sqrt{\frac{1}{M}\sum_{k=1}^{M}\left(T_k - \bar{T}\right)^2}$$

denote the mean and the standard deviation of the computational time T_k required to run an estimator over the Monte Carlo simulation. For PF with $N_p = 20000$, we have $\bar{T} = 2.74$ s and $\sigma_T = 1.27 \times 10^{-1}$ s. For RUKF with $c = 0.015$, we have $\bar{T} = 9.6 \times 10^{-3}$ s and $\sigma_T = 2.95 \times 10^{-3}$ s. For RCKF with $c = 0.015$, we have $\bar{T} = 9 \times 10^{-3}$ s and $\sigma_T = 2.12 \times 10^{-3}$ s. Figure 1 illustrates the performance of the estimators in terms of MSE_t where RUKF, CUKF, RUKF-a, CUKF-a are with $c = 0.015$ and PF is with $N_p = 20000$. We can notice that the performance of PF (black line) gradually deteriorates over time compared to that of the proposed robust filters (dashed green line and dashed orange line).

For completeness, we also consider the average mean squared prediction error over the time horizon:

$$\mathrm{MSE}^p_t = \frac{1}{M}\sum_{k=1}^{M}\left\|x^k_t - \hat{x}^k_t\right\|^2, \tag{19}$$

where $\hat{x}^k_t$ denotes the one-step-ahead prediction of x^k_t given $y^k_0, \dots, y^k_{t-1}$. Moreover, $\overline{\mathrm{MSE}}^p$ denotes the average of MSE^p_t over the time horizon. Table II reports the index $\overline{\mathrm{MSE}}^p$ for the filters, where the robust filters are evaluated for different values of the tolerance c, and PF is evaluated for different numbers of particles N_p. Figure 2 shows the performance

TABLE II: $\overline{\mathrm{MSE}}$ for the different estimators, with the grey cells highlighting the best performance.

c	RUKF (UKF)	RUKF-a (UKF-a)	RCKF (CKF)	RCKF-a (CKF-a)	N_p	PF
0	0.415	0.466	0.420	0.468	100	0.567
0.0001	0.411	0.464	0.415	0.468	500	0.456
0.001	0.400	0.490	0.404	0.498	1000	0.455
0.005	0.382	0.502	0.386	0.507	2000	0.403
0.01	0.371	0.504	0.374	0.500	3000	0.409
0.015	0.366	0.519	0.367	0.588	5000	0.390
0.02	0.367	0.535	0.365	0.593	8000	0.391
0.03	0.415	0.654	0.371	0.658	10000	0.378
0.05	0.608	0.891	0.455	0.918	20000	0.365

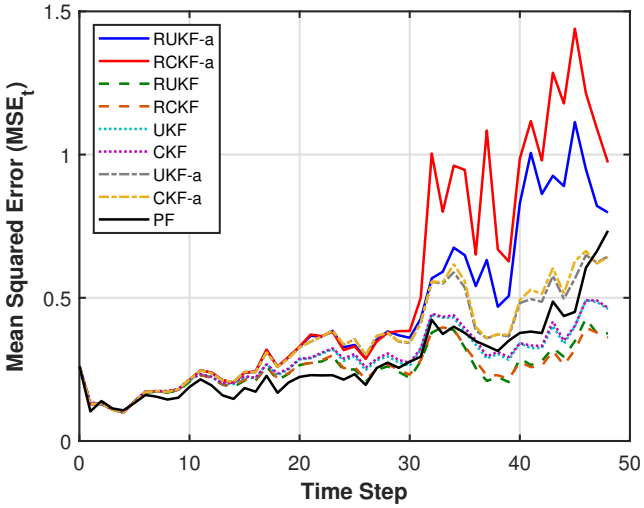


Fig. 1: Performance of the filters in terms of MSE_t; RUKF, RCKF, RUKF-a, RCKF-a are with $c = 0.015$; PF is with $N_p = 20000$.

of the estimators in terms of MSE^p_t, where RUKF, CUKF, RUKF-a, and CUKF-a are evaluated with $c = 0.015$, and PF is evaluated with $N_p = 20000$. The observations and comments are consistent with those made in the previous case.

In the previous Monte Carlo simulation, it appeared that the PF could provide a performance similar to that of the RUKF and RCKF, at the cost of using a large number of particles N_p. However, this is not true in general. In particular, this becomes evident when the noises v_t and w_t are less relevant in the model dynamics, i.e. when the covariance matrices Q and R are reduced toward zero. More precisely, we consider the same Monte Carlo simulation as before, but with $Q = 0.1I_2$ and $R = 0.1$. Table III reports the index $\overline{\mathrm{MSE}}$ for the RUKF and RCKF (for different values of c) and for PF (for different numbers of particles N_p). As we can see, PF performs remarkably worse than RUKF and RCKF.

V. Conclusion

In this paper, we have extended the robust sigma-point filtering framework proposed in [25] to state-space models with non-additive noise. Specifically, we have characterized

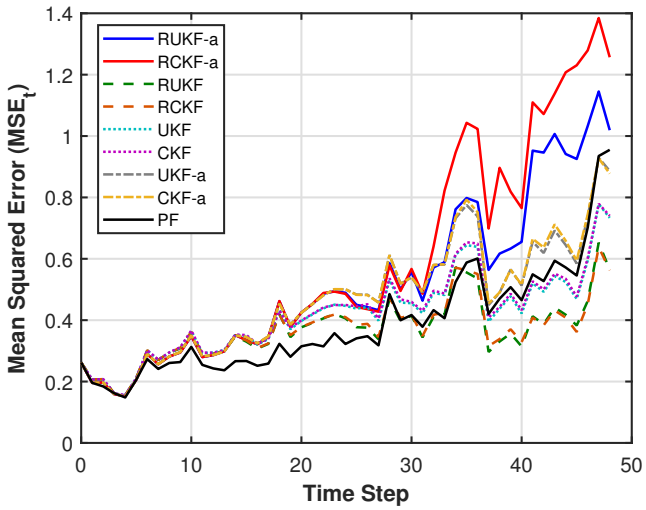


Fig. 2: Performance of the filters in terms of MSE_t^p; RUKF, RCKF, RUKF-a, RCKF-a are with $c = 0.015$; PF is with $N_p = 20000$.

TABLE III: $\overline{\text{MSE}}$ for the different estimators in the second Monte Carlo simulation, with the grey cells highlighting the best performance.

c	RUKF	RCKF	N_p	PF
	(UKF)	(CKF)		
0	0.074	0.074	100	0.274
0.0001	0.078	0.078	500	0.203
0.001	0.071	0.071	1000	0.181
0.01	0.051	0.052	2000	0.164
0.05	0.032	0.032	3000	0.151
0.08	0.030	0.031	5000	0.142
0.1	0.035	0.050	10000	0.132
0.15	0.128	0.065	20000	0.122

the robust sigma-point estimator as the solution to an approximate minimax game, where the estimator minimizes the variance of the estimation error, while the adversary maximizes it by selecting the least favorable model within an ambiguity set. Simulation results demonstrate the necessity of adopting the proposed strategy to effectively handle the case of state space models with non-additive noise.

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REFERENCES

[1] S. Yi and M. Zorzi, "A robust approach to sigma point kalman filtering," *arXiv*, vol. 2506.04815, 2025.
[2] S. Yi, X. Jin, Z. Wang, Z. Liu, and M. Zorzi, "Data-driven robust uav position estimation in gps signal-challenged environment," in *2025 American Control Conference (ACC)*, pp. 491–496, 2025.
[3] T. D. Barfoot, *State estimation for robotics*. Cambridge University Press, 2024.
[4] G. Y. Kulikov and M. V. Kulikova, "Hyperbolic-SVD-based square-root unscented Kalman filters in continuous-discrete target tracking scenarios," *IEEE Transactions on Automatic Control*, vol. 67, no. 1, pp. 366–373, 2022.
[5] J. Ajgl and O. Straka, "Covariance intersection in track-to-track fusion with memory," in *2016 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pp. 359–364, 2016.
[6] M. Havlicek, K. J. Friston, J. Jan, M. Brazdil, and V. D. Calhoun, "Dynamic modeling of neuronal responses in fMRI using cubature Kalman filtering," *NeuroImage*, vol. 56, no. 4, pp. 2109–2128, 2011.
[7] M. I. Ribeiro, "Kalman and extended kalman filters: Concept, derivation and properties," *Institute for Systems and Robotics*, vol. 43, no. 46, pp. 3736–3741, 2004.
[8] J. Prüher, T. Karvonen, C. J. Oates, O. Straka, and S. Särkkä, "Improved calibration of numerical integration error in sigma-point filters," *IEEE Transactions on Automatic Control*, vol. 66, no. 3, pp. 1286–1292, 2021.
[9] J. Prüher and O. Straka, "Gaussian process quadrature moment transform," *IEEE Transactions on Automatic Control*, vol. 63, no. 9, pp. 2844–2854, 2018.
[10] E. A. Wan and R. Van Der Merwe, "The unscented Kalman filter for nonlinear estimation," in *Proceedings of the IEEE 2000 Adaptive Systems for Signal Processing, Communications, and Control Symposium (Cat. No. 00EX373)*, pp. 153–158, IEEE, 2000.
[11] I. Arasaratnam and S. Haykin, "Cubature Kalman filters," *IEEE Transactions on Automatic Control*, vol. 54, no. 6, pp. 1254–1269, 2009.
[12] S. Särkkä and L. Svensson, *Bayesian filtering and smoothing*, vol. 17. Cambridge university press, 2023.
[13] Y. Huang, Y. Zhang, Z. Wu, N. Li, and J. Chambers, "A novel adaptive kalman filter with inaccurate process and measurement noise covariance matrices," *IEEE Transactions on Automatic Control*, vol. 63, no. 2, pp. 594–601, 2018.
[14] Y. Huang, Y. Zhang, Y. Zhao, P. Shi, and J. A. Chambers, "A novel outlier-robust kalman filtering framework based on statistical similarity measure," *IEEE Transactions on Automatic Control*, vol. 66, no. 6, pp. 2677–2692, 2021.
[15] L. P. Hansen and T. J. Sargent, *Robustness*. Princeton university press, 2008.
[16] R. Boel, M. James, and I. Petersen, "Robustness and risk-sensitive filtering," *IEEE Transactions on Automatic Control*, vol. 47, no. 3, pp. 451–461, 2002.
[17] L. P. Hansen, T. J. Sargent, *et al.*, "Robust estimation and control under commitment," *Journal of economic Theory*, vol. 124, no. 2, pp. 258–301, 2005.
[18] B. Levy and R. Nikoukhah, "Robust state-space filtering under incremental model perturbations subject to a relative entropy tolerance," *IEEE Transactions on Automatic Control*, vol. 58, pp. 682–695, Mar. 2013.
[19] S. Yi and M. Zorzi, "Robust Kalman filtering under model uncertainty: The case of degenerate densities," *IEEE Transactions on Automatic Control*, vol. 67, no. 7, pp. 3458–3471, 2021.
[20] M. Zorzi, "Convergence analysis of a family of robust kalman filters based on the contraction principle," *SIAM Journal on Control and Optimization*, vol. 55, no. 5, pp. 3116–3131, 2017.
[21] M. Zorzi, "On the robustness of the Bayes and Wiener estimators under model uncertainty," *Automatica*, vol. 83, pp. 133–140, 2017.
[22] O. Straka, J. Duník, and V. Elvira, "Importance gauss-hermite gaussian filter for models with non-additive non-gaussian noises," in *2021 IEEE 24th International Conference on Information Fusion (FUSION)*, pp. 1–7, 2021.
[23] A. M. Maras and E. A. Kokkinos, "Space—time threshold detection in non-additive non-gaussian noise fields," *Journal of the Franklin Institute*, vol. 336, no. 4, pp. 675–686, 1999.
[24] S. Yi and M. Zorzi, "An update-resilient Kalman filtering approach," *Automatica, provisionally accepted as regular paper*, 2025.
[25] B. Levy and R. Nikoukhah, "Robust least-squares estimation with a relative entropy constraint," *IEEE Transactions on Information Theory*, vol. 50, no. 1, pp. 89–104, 2004.